INCH-POUND

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27 October 1989
SUPERSEDING
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8 December 1986

MILITARY SPECIFICATION

TRUCKS, LIFT, FORK, ELECTRIC MOTOR DRIVEN,

2000-3000-4000-6000 POUND CAPACITY

GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirement for battery powered, industrial, depot type forklift trucks.

1.2 Classification. Forklifts trucks shall be of the following types as specified.

a. Type E - For general warehouse use.

b. Type EE - For handling of ammunition and explosives.

c. Type EX - For atmospheres having flammable vapors and dust.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN : STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3930

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SPECIFICATIONS

FEDERAL

GG-P-455	- Plates and Foils, Photographic (Photosensitive Anodized Aluminum).

Military

MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Material.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-L-2105	- Lubricating Oil, Gear, Multi Purpose Metric.
MIL-W-5044	- Walkway Compound, Nonslip, and Walkway Matting, Nonslip.
MIL-W-5050	- Walkway Coating and Matting, Nonslip, Aircraft, Application of.
MIL-M-7866	- Molybdenum Disulfide, Technical Lubrication Grade.
MIL-G-10924	- Grease Automotive and Artillery.
MIL-S-43926	- Suit, Chemical Protective.
MIL-G-43976	- Gloves and Glove Set, Chemical Protective.
MIL-F-43987	- Footwear Cover, Chemical Protective (Overboots).
MIL-R-46164	- Rustproofing for Military Vehicles and Trailers.
MIL-C-46168	- Coating Aliphatic, Polyurethane, Chemical Agent Resistant.
MIL-B-46176	- Brake Fluid, silicone, Automotive, All Weather, Operational, and Preservative Metric.
MIL-M-51282	- Mask Chemical, Biological Field M17A1.
MIL-H-51291	- Hood Chemical, Biological Mask M6A2.
MIL-C-62218	- Corrosion Preventive Compounds, Cold Application.

(see supplement 1 for list of associated specifications)

STANDARDS

FEDERAL

FED-STD-595	- Colors.

MILITARY

MIL-STD-162	- Materials Handling Equipment: Preparation for Shipment, Storage, Cyclic Maintenance, Routine Testing and Processing.

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MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1474	- Noise Limits for Army Materiel.
MIL-STD-1791	- Designing for Internal Aerial Delivery in Fixed Wing Aircraft.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government, documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in the solicitation.

DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION

29 CFR, Chapter XVII, Part 1910
Occupational Safety and Health Standards

(Applications for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

MILITARY TRAFFIC MANAGEMENT COMAND (MTMC/TEA)

Phamphlet 70-1 - Transportability for Better Strategic Mobility.

(Application for copies should be addressed to the Military Traffic Management Command, Transportation Engineering Agency (MTMC/TEA), ATTN: MTT-TR, P.O. Box 6276, Newport News, VA 23606-0276)

2.2 Non-Government publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues cited in the solicitation (see 6.2).

THE AMERICAN ASSOCIATION OF MECHANICAL ENGINEERS (ASME)

ASME/ANSI B 56.1	- Safety Standard for Low Lift and High Lift Trucks.
ASME/ANSI B 56.11.3	- Load Handling Symbols for Powered Industrial Trucks.
ASME/ANSI B 56.11.4	- Hook-Type Forks and Fork Carriers for Powered Industrial Forklift Trucks.

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(Application for copies should be addressed to the American Association of Mechanical Engineers (ASME) 345 East 47th Street, New York, NY 10017).

NATIONAL ELECTRICAL MANUFACTURERS ASS0CIATION (NEMA)

NEMA Standards.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street NW, Washington DC 20037.)

NATIONAL MOTOR FREIGHT TRAFIC ASSOCITION (NMFTA)

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, ATTN: Traffic Order Section, 2200 Mill Road, Alexandria, VA 22314).

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

J209	- Instrument Face Design and Location for Construction and Industrial Equipment.
J297	- Operator Controls on Industrial Equipment.
J514	- Hydraulic Tube Fittings.
J516	- Hydraulic Hose Fittings.
J517	- Hydraulic Hose.
J518	- Hydraulic Flanged Tube, Pipe, and Hose Connections, 4-Bolt Split Flange Type.
J524	- Seamless Low Carbon Steel Tubing Annealed for Bending and Flaring.
J525	- Welded and Cold Drawn Low Carbon Steel Tubing Annealed for Bending and Flaring.
J534	- Lubrication Fittings.
J753	- Lubrication Chart-Construction and Industrial Machinery.
J833	- USA Human Physical Dimensions.
J898	- Control Locations for Off-Road Work Machines.
J899	- Operator's Seat Dimensions for Off-Road Self-Propelled Work machines.
J925	- Minimum Service Access Dimension for Off-Road Machines.
J931	- Hydraulic Power Circuit Filtration.
J1176	- External Leakage Classifications for Hydraulic Systems.
J1362	- Symbols for Controls, Indicators, and Tell-Tales for Off-Road, Self-Propelled Work Machines.

(Applications for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATTON, INC. (TRA)

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Tire and Rim Association Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

UNDERWRITERS LABORATORIES INC. (UL)

ANSI/UL 583 - Standard for Electric-Battery-Powered Industrial Trucks

(Application for copies should be addressed to Underwriters Laboratories, Inc., Publications Stock, 333 Pfingsten Road, Northbrook, Illinois 60062.)

UNIFORM CLASSIFICATION COMMITTEE (UCC).

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The truck shall be a sit-down rider 36-volt system, electric powered counterbalanced forklift truck, equipped with a battery and electric drive motor silicon controlled rectifier (SCR) control, a battery discharge indicator, a high visibility roller mast, and as specified herein. The forklift shall be the latest model of the manufacture's standard commercial product. The introduction of normal product improvement changes in the one year period preceding the date of solicitation for procurement is acceptable. The forklift shall be equipped with standard instruments, components, and accessories normally required for the safe and effective operation of the truck. The truck shall be complete with all components that are standard with the contractor's products, whether stipulated herein or not, together with such accessories, and options, as may be specified herein. The truck shall be equipped with all other components and parts not specifically mentioned but necessary to provide a functional machine and shall conform in quality to that normally provided to the commercial industry.

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TYPE	DESCRIPTION
E	Electric, minimum safeguards.
EE	Electric motors and all other equipment completely enclosed.
EX	Electrical fittings and equipment designed, constructed, and assembled that the units may be used in certain atmospheres containing flammable vapors and dusts.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (6.3) in accordance with 4.3.

3.2 Specification sheet. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheets, the latter shall govern.

3.3 Recovered material. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the forklift may be newly fabricated from recovered materials to the maximum extent practicable, provided the forklift produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the forklift.

3.3.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.3 Material deterioration prevention and control. The forklift shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be in any of the applicable operating and storage environments to which the forklift may be exposed.

3.4 Transportability. The trucks shall be capable of being transported by military or commercial trailers, trains, marine vessels, aircraft and must be capable of withstanding the impact forces encountered in shipment, without damage or permanent deformation. The trucks shall be equipped with tiedown or slinging provisions.

3.4.1 Tiedown provisions. The tiedown provisions shall conform to MIL-STD-209, class 2 or 3, type II equipment, and to MIL-STD-1791 for equipment restraint criteria. The tiedown provisions shall satisfactorily complete the pull testing as specified without weld failure, permanent deformation,

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cracking, loosening, or breaking of the provision or its connecting structural components.

3.4.2 Slinging provisions. The slinging provisions shall conform to MIL-STD-209, class 2 or 3, type II. The provisions shall enable the complete truck to be lifted in the normal operating position. The provisions shall be located so that not less than one inch clearance is maintained between slings and all exterior parts and shall be fastened to members which will withstand stresses in the amount and direction of pull specified for the provisions without weld failure, permanent deformation, cracking, loosening, or breaking of the provision or its connecting structural components. Slinging provisions may also be used as tiedown provisions when such provisions meet the requirements specified in 3.4.1. All slinging/ tiedown provisions shall be labeled 'LTFT', 'TIEDOWN' or "LIFT TIEDOWN", as appropriate.

3.4.3 Air transportability. The trucks shall meet the requirements of MIL-STD-1791 for air transport. When required (see 6.2), the contractor shall provide sufficient data to the Government in the Technical Information Package to confirm that the trucks being supplied can be loaded, transported in the C-130, C-141, C-5A aircrafts and lifted by helicopters.

3.4.4 Rail transportability. The trucks shall be rail transportable in CONUS and NATO countries without restrictions. The trucks shall have a dimensional profile within the Gabarit International de Chargement (GIC) in accordance with MIMC/TEA pamphlet 70-1, outline diagram when loaded on a 50-inch (127 cm) high rail car. The trucks shall be capable of withstanding shock loads resulting from rail impact testing in accordance with 4.4.18 without failure, damage, or permanent deformation.

3.4.5 Highway transportability. The trucks when loaded on a semitrailer/tractor, shall be within the highway permit limits of all States.

3.5 Environmental requirements. The truck shall operate in any ambient temperature from 0 ℉ to +110 ℉ when tested in accordance with 4.4.10.

3.6 Lubrication. Lubrication means shall be provided for all moving parts that require lubrication, except where sealed permanent lubrication is provided. Each forklift shall be serviced and run-in with military lubricants conforming to MIL-L-2104, MIL-L-2105, MIL-B-46176, MIL-M-7866, and MIL-G-10924. Initial servicing and run-in may be accomplished with lubricants standard to the contractor's commercial practice (except for braking system) if compatible with above military lubricants. (If commercial lubricants are utilized during testing, the manufacturer will provide certification from suppliers that commercial lubricants are compatible with military lubricants prior to the start of the first article test). A lubrication data plate in accordance with SAE J753 shall be mounted on the forklift and shall identify all military lubricants and appropriate temperature ranges to be utilized when servicing the forklift. The data shall conform to MIL-P-514, type III, composition C (GG-P-455), type I, grade A, class 1. Grease lube fittings shall conform to SAE J534.

3.7 Maintainability. Provision shall be made for adjustment, servicing, and replacement of all electrical assemblies and components, hydraulic system

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components or assembly where applicable, battery, wearing parts of lift and tilt mechanism, brakes and components, wear parts of steering assembly, tires, wheels, lights, and horn, without the removal of any major parts other than covers or requiring special tools. If hand access openings are used on the exterior of the truck, the edges of each opening shall be smooth and the opening shall be provided with a removable or hinged cover, whenever it is required. Covers shall be provided with captive, hand operable fasteners. Minimum service access openings shall be in accordance with SAE J925. There shall be no interference with the servicing or draining of oils and lubricants to or from any assembly or components by frame members or other obstructions. Drain holes with removable drain plugs shall provide for complete drainage without disconnection of hoses. Drain outlets shall provide for drainage into a receptacle. All fuses/circuit breakers shall be readily accessible without removal of other components.

3.8 Maintenance operations. Unless otherwise specified herein, each operation in the following maintenance operations list shall be accomplished by one man in not more than the tire specified, using common tools and special tools, if any, furnished with the truck.

Time in minutes	Type E	Type EE	Type EX
a. Remove and replace hydraulic motor brushes.	60	90	150
b. Remove and replace steering motor brushes.	60	90	150
c. Remove and replace drive motor brushes.	60	90	150
d. Remove and replace all contractor tips.	60	90	150
e. Remove and replace all filters screens and strainers in hydraulic system.	105	105	105
f. Bleed and adjust brakes and refill master cylinder.	45	45	45
g. Remove and replace floodlight and tail light bulbs (not including black out bulbs).	30	30	45
h. Lubricate all lubrication fittings with 2 strokes per fitting using grease gun and extension.	60	60	60

3.9 Hydraulic system. The hydraulic system shall consist of all hydraulic components necessary for operation of the load handling means system and shall not include break operations. All hydraulic system component material shall be compatible with MIL-L-2104. The system shall provide for lowering of capacity load at a speed not to exceed 80 FPM in the event of or damage to hydraulic hose(s) supplying the lift cylinder. A pipe tap or straight thread O-ring test point for measuring pressure shall be located at the outlet of each pump or pressure inlet of each control valve. Hydraulic hose(s) shall have a working pressure which is greater than the hydraulic system maximum relief valve setting. A system filter shall be furnished in either the suction or return line. The system filter shall meet the requirement of SAE J 931a. Hose reels shall be mounted such that they are within the plan outline of the truck. Positive means shall be furnished to prevent cavitation of the lift cylinder.

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3.9.1 Hydraulic tubing, hoses and fittings. All hydraulic tubing, hoses and fittings shall meet the requirement of the applicable SAE recommended practices paragraphs, J 514, J 516, J 517, J 518, J 524, and J 525, and shall be routed to prevent kinking, chafing, and twisting. All lines and fittings shall be tagged or identified in some way.

3.9.2 External leakage. After completing test specified in 4.3, the degree of external hydraulic system leakage shall not exceed a class 4D for dusty conditions of SAE J 1176. External hydraulic system leakage shall not exceed a class 3 for production trucks under normal conditions.

3.10 Electrical equipment.

3.10.1 Static discharge device. When type EE specified the truck shall be equipped with two replaceable electrically conductive rubber straps, mounted immediately adjacent to the wheels.

3.10.2 Safety cutout system. Provisions shall be made to disrupt power to the traction motor when the operator leaves his seat and when the operator applies the service brakes.

3.11 Electric motors. Motors shall be industrial type with a high starting torque. The insulation for motors shall be class B, F, or H as defined in NEMA standards. When other than type E truck is specified (see 6.2), one or more thermal relay switches shall be provided which shall cause disconnection of battery power to the affected circuit before the outside surface temperature of the motor(s) exceeds 347 °F. Thermal relays shall be located in critical heat areas and shall interrupt the current only to the over-heated circuit.

3.12 Uprights and carriage assembly. Uprights shall be the telescopic roller type. Main load rollers shall be the permanently lubricated for life type. The forks and fork carrier shall conform to ASME/ANSI B 56.1 and ASME/ANSI B 56.11.4. The fork carrier shall be free of obstructions to allow the forks to slide across the entire width of the carrier. A cutout shall be provided in the fork carrier so that the lower hook of the fork will fit through the slot and each fork shall be removable without removing the load backrest. Fork thickness shall not exceed 0.5 inch at the tips and the end taper shall be not less than 14 inches long. Fork size shall be as specified on the applicable specification sheet. Forks shall be adjustable to obtain the fork spacing measured from the outside of the forks as specified on the applicable specification sheet.

3.12.1 Load backrest. A metal backrest, removable without disturbing any other component, shall be provided. There shall be no protruding bolts or appendages beyond the side plane of the load backrest. In conjunction with the fork and hanger design, it shall provide a vertical rear guard at least 48 maximum high, measured from the load carrying surface of the forks, and at least equal to the width of the carriage.

3.13 Wheels and tires. Tire type shall be as specified on the applicable specification sheet.

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3.13.1 Wheels for solid tires. The truck shall be mounted on steel, malleable iron, or cast iron wheels, equipped with tapered roller bearings.

3.13.1.1 Solid tires. The tires shall be row and unused, standard solid or cushion style, and industrial type. Front tires shall have commercial, non-directional type tread; rear tires may have either smooth rib, or non-directional type tread. Tire loadings shall not exceed the values at speeds specified herein listed in the Tire and Rim Yearbook.

3.13.2 Pneumatic tires. Tire loadings shall not exceed the values at speeds, specified herein listed in the Tire and Rim Yearbook. When specified (see 6.2), the truck shall be equipped with solid filled puncture proof tires. Tires shall be not more than 12 months old on date of acceptance by the Government.

3.14 System safety. The trucks shall conform to the applicable requirements of ASME/ANSI B 56.1, ANSI/UL 583, ASME/ANSI B 56.11.3 and OSHA in effect at the time of manufacturer and other requirements specified herein. Unless otherwise specified, trucks shall be type E (see 6.2). When specified (see 6.2) individual trucks shall be affixed with safety rating labels issued by independent testing laboratories. Asbestos shall not be used in the manufacturing of this vehicle.

3.15 Steering. Power steering shall be furnished for all trucks having a rated capacity greater than 2000 pounds. Steering shall not require more than 5 lbs force. In the event of power failure, steering shall not require more than 35 lbs force.

3.16 Towing device. A pintle or pin-type towing device shall be provided in the center rear of the truck.

3.17. Brakes. Hydraulically actuated foot brakes shall be provided. Brakes shall be self adjusting. Application of brakes shall require not less than 5 pounds or more than 15 pounds of pedal force. In the event of brake power assist failure pedal force shall not exceed 75 pounds. Brakes shall meet or exceed requirements of ANSI B 56.1. The brake pedal shall not exceed an angle of 60 degrees to the horizontal and shall be located so that the operator can reach it without extreme stretching when the brake is fully applied. The brake pedal shall be located a minimum distance of 4 inches from any other pedal or control.

3.17.1 Parking brake. A seat operated parking brake shall be provided. The brake shall be automatically activated when the operator leaves the operator's seat and hold the capacity loaded vehicle on the specified slope. The braking capacity shall conform to ASME/ANSI B 56.1.

3.17.2 Brake fluid. Brake system shall be compatible with MIL-B-46176 brake fluid, silicone, automotive all weather operational and preservation. The manufacturer shall provide in the first article test report, certifications from manufacturers of brake components, that the brake components utilized within the forklift brake system are compatible with silicone brake fluid. A tag shall be attached to the brake master cylinder stating "Caution." Use Silicone Brake Fluid Only MIL-B-46176.

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3.18 Seat. The seat shall conform to the requirements of SAE J 899.

3.19 Battery. The truck shall be furnished with battery(s) having a capacity of not less than that specified on the applicable specification sheet. Unless otherwise specified (see 6.2), batteries will be shipped wet and charged.

3.19.1 Battery component. The battery compartment shall be provided with a pictorial diagram showing battery connections which are visible when the compartment is open for maintenance. The batteries shall be secured inside the compartment to prevent unnecessary movement. The batteries shall be sufficiently insulated to prevent accidental high current short and shall be readily accessible for servicing. The compartment shall have holes in the bottom to allow for drainage of excess fluids.

3.19.2 Batter connector. Unless otherwise specified (see 6.2), the truck shall be equipped with SB type connectors.

3.19.3 Battery warning. A warning sign shall be affixed to the exterior of the battery compartment that states the hazards present with the use of batteries.

3.20 Human factors engineering.

3.20.1 Controls and instrumentation. All controls and instrumentation shall be provided with position markings conforming to ASME/ANSI B 56.11.3 and labeled in accordance with SAE J 1362. The controls shall be operable and the instruments shall be viewable by the 5th to the 95th percentile male and female, according to the dimensions in SAE J 833a and shall be in accordance with SAE J 297 and SAE J 898. Instrumentation shall be in accordance with SAE J 209.

3.20.1.1 Discharge indicator. A battery discharge indicator containing a lift interrupt shall be provided.

3.20.1.2 On-off switch. A key-operated on-off switch shall be furnished. It shall have not less than two positions. The on and off positions shall be labeled "ON" and "OFF" respectively. Two keys shall be furnished with each truck. All keys shall be made to fit each and every truck.

3.20.1.3 Hour meter. An electrically operated hour meter shall be provided, which registers up to 9999 hours in 0 .1 hour increments. The hour meter shall be accessible for reading without removing any component of the truck.

3.20.2 Human factors. The forklift shall be operable and maintainable by the full range of user personnel (5th percentile female through 95th percentile male) in accordance with, but not necessarily limited to, the requirements of SAE J209, J297, J298, J833, J898, or ASM/ANSI B 56.11.3.

3.20.3 Nuclear-biological-chemical (NBC) protection. When specified (see 6.2), the forklift shall be operable and maintainable by the full range of the

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user personnel 5th percentile female through 95th percentile male) wearing MOPP IV protective garments, i.e.;

a. Mask, chemical, biological, field M17A2 MIL-M-51282
b. Hood, chemical, biological, mask, M6A2 MIL-H-51291
c. Suit, chemical, protective, MIL-S-43926
d. Glove and glove set, chemical protective, MIL-G-43976
e. Footwear cover, chemical, protective (overboots), MIL-F-43987

The operator shall be capable of activating all controls (i.e., levers, switches, pedals, knobs, and handles) without interference caused by the ensemble that is detrimental to the operation of the forklift.

3.21 Lights. Each truck shall be equipped with one sealed beam floodlamp mounted at the front of the truck, one automotive type combination stoplight and tail-lamp mounted at the rear and instrument panel illumination. The flood-lamp shall be shock mounted in an elastomer ring housing on the left side of the mast. Directional manual adjustment range of this flood-lamp shall be a minimum of 45 degrees above and below the horizontal plane. Horizontal adjustment range of the flood-lamp shall be a minimum of 300 degrees. Flood-lamp shall be protected against damage by a means of position or a guard.

3.22 Side shift. Unless otherwise specified (see 6.2) truck shall be equipped with a side shift mechanism. The mechanism shall be hydraulically operated to move the forks a minimum of 4 inches either side of center with rated capacity load. The sideshift mechanism shall not interfere with the ability to remove the forks from the fork hanger.

3.23 Operator's overhead guard. A removable overhead guard shall be furnished which meets applicable requirements of ASME/ANSI B 56.1. The guard shall be removable with handtools. No component shall be mounted on the overhead guard. Guard height, measured from ground level to the highest point of the guard, shall not exceed height specified on applicable specification sheet. Guard design shall not interfere with the operation of the truck or battery replacement nor with normal movements of the operator when entering, leaving, or operating the truck. The overhead guard shall remain firmly secured to its support brackets under all operating conditions specified herein.

3.24 Performance.

3.24.1 Lifting speed. Lift speed with capacity load on the forks shall be as is specified on the applicable specification sheet.

3.24.2 Lowering speed. Speed of lowering of unloaded forks shall be not less than 40 feet per minute average over the entire distance from maximum fork height to ground level. Speed of lowering forks with capacity load shall be not more than 80 feet per minute over the entire distance from maximum fork height to ground level.

3.24.3 Right angle turn. The truck, when carrying capacity load, shall be capable of backing through a turn in either direction and returning to the

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same or parallel position within dimension specified on the applicable specification sheet, when operated as specified herein.

3.24.4 Speed. While carrying the rated capacity load on a level surface, truck shall be capable of attaining speeds specified on the applicable specification sheet in both the forward and reverse directions.

3.24.5 Slop ascension, first speed, forward direction. The truck shall be capable of ascending a 15 percent grade on a dry concrete surface with and without capacity load. The truck shall be able to accelerate from dead stop on this slope when carrying capacity load.

3.24.6 Upright tilt. When unloaded, truck mast shall have the forward tilt and rearward tilt specified on the applicable specification sheet. Positive means shall be furnished to prevent cavitation of the tilt cylinders.

3.24.7 Collapsed mast height. The collapsed mast height of an unloaded truck measured from the ground to the top of the uprights in a vertical position shall not exceed that specified in the applicable specification sheet.

3.24.8 Maximum fork height. The maximum fork height of a truck bearing capacity load with the uprights fully extended, measured from the ground to the load carrying surface of the forks at the capacity load center, shall be not less than that specified in the applicable specification sheet.

3.24.9 Free lift height. Height of the forks without increasing the specified collapsed mast height shall be not less than that specified in the applicable specification sheet.

3.24.10 Stability. The truck shall conform to "forward stacking", "forward travel", "lateral stacking", and "lateral travel" stability requirements of ANSI B 56.1.

3.24.11 Endurance/reliability. The truck(s) shall complete 4000 cycles over a course as shown in figure 1 in accordance with test method 1. One endurance failure or a total of four reliability failures will be cause for rejection. Failures are as defined in 4.4.10.

3.24.12 Underclearance. The truck with capacity load shall have sufficient underclearance to permit operation from one horizontal plane to another up or down a 15 percent slope and over the obstacle course described in test method no. 1. Minimum underclearance beneath the mast assembly, when in true vertical position, shall be not less than that specified on the applicable specification sheet when the truck is loaded.

3.24.13 Drift. The lift assembly shall be capable of holding the rated load at maximum height for not less than 2 minutes with not more than 1 inch of vertical drift and not more than 0.5 degree of rotational drift from the vertical. Specified drift rates shall be measured when hydraulic oil temperature is not less than 120 degrees fahrenheit.

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3.25 Treatment and paint. The parts of the truck normally painted shall be cleaned, treated, and painted in accordance with the contractor's commercial practice. Unless otherwise specified (see 6.2) final color shall be contractor's normal color. The cowl overhead guard and mast may be matt black in color. When specified (see 6.2), the final color shall be in accordance with color 13538 of FED-STD-595 and treated and painted in accordance with MIL-T-704, type A (also see 3.3.4.5, color in special purpose vehicles and equipment and footnote 1 to table 1I in MIL-T-704). When specified (see 6.2), coating aliphatic, polyurethane, chemical agent resistant (CARC) MIL-C-46168 shall be used. However CARC shall not be applied to surfaces which exceed 400 °F. An alternate high temperature paint, in the appropriate color shall be used. When CARC is specified, the identification plate (see 3.27) shall state "Painted with CARC". The vehicle shall be rustproofed as per the instructions provided in MIL-R-46164 and MIL-C-62218.

3.25.1 Walkway coating. Floor plates and step surfaces shall have skid resistant surfaces conforming to MIL-W-5044 and MIL-W-5050.

3.26 Identification marking. Each truck shall be identified with an identification plate conforming to MIL-P-514, type I, style 1, composition C, of type I, grade A, class 1 material which shall be located in a visible and permanent location on the truck. The identifying data shall be stamped in letters in accordance with MIL-P-514. The plate shall be securely attached to the truck with screws, bolts, rivets, or by welding and shall be furnished and mounted by the contractor.

3.26.1 Truck marking. Marking on trucks shall be painted black or other designated color, using the designated military paint specified in 3.25. Characters shall be block type capitals, and Arabic numerals. Location and character size shall be as follows:

Information	Example	Location	Minimum character size (Ht.)
Agency	US	On each side and rear	1.5 inches
Registration number	10D315	On each side and rear	1.5 inches
Capacity	6000lb. Cap	On each side	2 inches
Safety	No smoking	On rear of mast	1 inch
	No riders	On rear of mast	1 inch
Tire pressure	80 psi	On truck above each wheel	1 inch
Lifting/tiedown points	Lift/tiedown	Near lift/tiedown points	1 inch

3.27 High altitude electromagnetic pules (EMP). When specified (see 6.2), schematic and electrical diagrams detailed to the component level (i.e., resistors, capacitors, filters, transistors, SCRs) with value of all components of the forklift shall be furnished by the contractor to the Government to assist the Government in performing the EMP evaluation. When specified (see 6.2), a forklift shall be furnished to the Government and shall be evaluated by the Government to determine necessary protection devices for protection against EMP.

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3.28 Government-loaned equipment. Unless otherwise specified (see 6.2), the following items shall be loaned by the Government (see 6.4)

a. NBC protective ensemble.

1. Chemical-biological mask, NSN 4240-00-926-4200.
2. Chemical-biological hood, NSN 4240-00-999-0420.
3. Chemical protective suit, NSN 8415-00-407-1062.
4. Chemical protective glove set, NSN 8415-01-0033-3519.
5. Chemical protective footwear, NSN 8430-01-021-5978.

3.29 Identical items. All forklifts furnished under this specification shall be physically and mechanically identical to the first article forklift that is approved by the Government.

3.30 Noise limits. The noise produced by the truck shall conform with MIL-STD-1474, requirements. The provisions of MIL-STD-1474, 4.3 and 4.4, shall be provided if and only if MIL-STD-1474, 5.1.1.2 procedures have been pursued and documented to the satisfaction of the procuring activity and written permission to exceed the 85 dB(A) limit is obtained from the procuring activity. Hazard signs shall conform with MIL-STD-1474 except the signs shall state "Wearing Protection Required within (Specify) Feet." The signs shall be readable at the distance indicated on the sign.

3.31 Electromagnetic interference. When specified (see 6.2), the electromagnetic interference emission characteristics of the forklift shall meet the limits specified in MIL-STD-461, part 8, class C1, group II, except that a 20 dB relaxation is permitted in the emission limits. The operators horn is exempt from electromagnetic interference requirements.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services Conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Classification of inspections, evaluations, and demonstrations. Inspections, evaluations, and demonstrations shall be classified as follows:

a. First article truck inspection (see 4.3).
b. Quality Conformance Inspection (see 4.5).
c. Inspection of packaging (see 4.9).

4.3 First article truck inspection. Prior to production the manufacturer shall furnish one or more trucks (see 6.2), produced from production tooling for first article examination and test. The trucks and their components shall be compared with this document, published literature, calculations and test data to verify compliance with this specification. The truck shall then be subjected to examination and test as marked in column 1, table I and column 1, table II. The presence of one or more defects shall constitute failure of the inspection. Prior to tests and examination, the following shall be performed:

a. Service the truck with oils and greases and a fully charged battery specified herein and designated for use in the ambient temperature at which the test shall be conducted.
b. Conduct run-in as prescribed by manufacturer.
c. Set the hydraulic system relief pressure as specified by manufacturer and record. Any change made to this shall constitute failure of all tests.
d. Furnish a schedule of maintenance to be followed during all testing of the first produced truck. A list of special tools required to perform this maintenance shall also be provided.
e. Electrical adjustments to set the battery discharge rate to be made prior to start of first article testing. Any changes made after starting these tests shall constitute a failure to all first article tests.

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TABLE I. Inspection schedule.

First article produced inspection	Quality conformance	Number	Defect	Requirement paragraph
X		101.	Recovered materials not identified.	3.3
X		102.	Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.	3.3.1
X		103.	Materials not as specified.	3.3
X		104.	Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration of the applicable storage and operating environments.	3.3.3
X	X	105.	System safety not as specified.	3.14
X	X	106.	Lubrication and fittings not as specified.	3.6
X		107.	Maintainability not as specified.	3.7
X	X	108.	Static discharge strap not as specified.	3.10.1
X	X	109.	Safety cutout system not specified.	3.10.2
X	X	110.	Electric motors not as specified.	3.11
X	X	111.	Hydraulic system not as specified.	3.9
X	X	112.	Hydraulic tubing hoses and fitting not as specified.	3.9.1
X		113.	External leakage not as specified.	3.9.2
X	X	114.	Uprights and carriage assembly not as specified.	3.12
X	X	115.	Load backrest not as specified.	3.12.1
X	X	116.	Wheels not as specified.	3.13, 3.13.1
X	X	117.	Tires not as specified.	3.13.1.1 3.13.1.2
X	X	118.	Steering not as specified.	3.15
X	X	119.	Brakes not as specified.	3.17
X	X	120.	Parking brake not as specified.	3.17.1
X	X	121.	Brake fluid not as specified.	3.17.2
X	X	122.	Towing device not as specified.	3.16
X	X	123.	Battery not as specified.	3.19
X	X	124.	Battery compartment not as specified.	3.19.1

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TABLE I. Inspection schedule. (Cont'd)

First article produced inspection	Quality conformance	Number	Defect	Requirement paragraph
X	X	125.	Battery connector not as specified.	3.19.2
X	X	126.	Battery warning sign not as specified.	3.19.3
X	X	127.	Controls and instrumentation not as specified.	3.20.1
X	X	128.	Discharge indicator not as specified.	3.20.1.1
X	X	129.	On-off switch riot as specified.	3.20.1.2
X	X	130.	Hour meter not as specified.	3.20.1.3
X	X	131.	Lights not as specified.	3.21
X	X	132.	Side shift not as specified.	3.22
X	X	133.	Overhead guard not as specified.	3.23
X	X	134.	Seat not as specified.	3.18
X	X	135.	Treatment and painting not as specified.	3.25
X	X	136.	Walkway coating not as specified.	3.25.1
X	X	137.	Identification marking not as specified.	3.26 3.26.1
X	X	138.	Transportability not as specified.	3.4
X	X	139.	Tiedown provisions not as specified.	3.4.1
X	X	140.	Slinging provisions not as specified.	3.4.2
X	X	141.	Air transportability requirements not met.	3.4.3
X	X	142.	Rail transportability requirements not met.	3.4.4
X	X	143.	Highway transportability requirements not met.	3.4.5
X	X	144.	Contractor does not have documentation available for identification of materials, material finishes, or	3.3.2
X	X	145.	treatment.	3.20.3
X	X	146.	Nuclear-biological-chemical protection not as specified.	5.3
X	X	147.	Preservation marking not as specified.	5.2
X	X	148.	Commercial preservation not as specified.	3.30
			Noise limits not as specified.	

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4.4 Test.

4.4.1 Test conditions. Test shall be conducted at the ambient temperature and climatic conditions existing at the place of test. Only that maintenance established by the contractor and submitted as a maintenance schedule prior to commencement of the test shall be performed during the test.

TABLE II. Test schedule.

First Article	Quality Conformance	Test	Test paragraph	Requirement paragraph
		First article truck inspection.	4.3	3.1.1, 3.6
X	X	Maximum fork height, collapsed mast height, free lift height.	4.4.3	3.24.7,3.24.8, 3.24.9
X	X	Lifting speed, lowering speed.	4.4.4	3.24.1, 3.24.2
X		Right angle turn.	4.4.5	3.24.3
X	X	Travel speed.	4.4.6	3.24.4
X		Slope ascension.	4.4.7	3.24.5
X		Tilt.	4.4.8	3.24.6
X		Stability.	4.4.9	3.24.10
X		Endurance reliability test.	4.4.10	3.24.11
X		Nuclear-biological-chemical protection.	4.7	3.20.3
X		Slinging and Tiedown.	4.4.15 4.4.16	3.4
X		Noise level test.	4.4.13	3.30
X		Maintenance operations test.	4.4.11	3.8
X		Brakes.	4.4.12	3.17
X		Drift	4.4.20	3.24.13
X		Electromagnetic interface	4.4.14	3.31

4.4.2 Capacity load. During the test specified herein, a 48-inch cube of the specified capacity load shall be used. Capacity load is defined as the truck capacity required by the applicable specification sheet. The tolerances for weight shall be plus 1 percent and minus 0 percent and for dimensions shall be plus 1 percent and minus 0 percent.

4.4.3 Maximum fork height, collapsed mast height, free-lift height. With capacity load on forks and mast in true vertical position measure and record free lift height and maximum fork height. Measure and record collapsed mast height with no load on the forks. Nonconformance to the applicable specification sheet shall be cause for rejection.

4.4.4 Lifting speed and lowering speed. Record time in seconds required to raise capacity load from floor to maximum fork height using timing device. Measure and record distance in inches from floor to top surface of forks in fully lowered position. Record the difference between maximum fork height and the height of the forks in the lowered position. Record time in seconds to lower unloaded forks from maximum height to fully lowered position. The raising and lowering shall be performed three times. Noncoformance to 3.24.1 or 3.24.2 shall be cause for rejection.

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4.4.5 Right angle turn. With capacity load on forks raised 6 inches above the ground and mast vertical demonstrate that truck can meet the right angle turn dimension as specified on the applicable specification sheet. Nonconformance shall be cause for rejection.

4.4.6 Travel speed. Operate the truck with capacity load in carry position. Truck shall attain speed specified in both directions. The truck shall be driven a distance necessary to attain maximum speed prior to entering a measured distance of not less than 22 feet. The truck shall be driven over the measured course at least three times in each direction. The speed shall be the average of the three runs in each direction. Inability of the truck to attain the minimum speed as specified on the applicable specification sheet shall be cause for rejection.

4.4.7 Slope ascension. Drive truck forward with capacity load along a horizontal approach and on to a ramp or slope of specified grade and bring to a complete stop halfway up the ramp. Start from a dead stop and proceed up the ramp to a horizontal surface. Repeat the procedure without the load. Nonconformance to 3.24.5 shall be cause for rejection.

4.4.8 Tilt. Place truck without load on flat level surface. Raise forks to an elevation of 2 feet and place mast in vertical position. Tilt mast to full forward position and record angle of tilt. Tilt mast to full rearward position and record angle of tilt. Nonconformance to 3.24.6 shall be cause for rejection.

4.4.9 Stability. Conduct "forward stacking", "forward travel", "lateral stacking", and "lateral travel" tilting platform tests in accordance with ASME/ANSI B 56.1. Nonconformance to ASME/ANSI B 56.1 shall be cause for rejection.

4.4.10 Endurance/reliability test. The truck carrying the capacity load, shall be operated for a minimum of 4000 cycles over a course as shown in figure 1 following the criteria laid out in test method 1. The surface shall be paved with concrete, asphalt, macadam or equivalent. Any incident resulting in internal component failure of the electric motor, differential, hydraulic pump, or permanent deformation of the chassis shall constitute an endurance failure. Fluid leakage, inability to perform operations or overheating shall also constitute an endurance failure. Any unscheduled maintenance action during this truck operation demonstration taking one person more than 15 minutes to correct shall constitute a reliability failure. Any endurance failure shall be cause for truck rejection. A total of 4 reliability failures shall be cause for rejection.

4.4.11 Maintenance operating test. Perform the maintenance operations as listed in 3.8. Nonconformance to 3.8, shall constitute failure of this test.

4.4.12 Braking test. Perform service brake test using the stopping distance method specified in ASME/ANSI B 56.1. Perform parking brake test specified in ASME/ANSI B 56.1 for sit down rider trucks. Nonconformance shall be cause for rejection.

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4.4.13 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 7. As a minimum, noise levels shall be measured when equipment is operating under full load, MIL-STD-1474, 5.1.2.1.4, contours shall be taken at not fewer than 12 equal (horizontal) arc increments. One increment shall include data from the noisiest position, additionally, the noise level at the typical operating position shall be provided as dB (A) level. Failure to comply with MIL-STD-1474 provisions shall constitute failure of this test.

4.4.14 Electromagnetic interference. The truck shall be tested in accordance with MIL-STD-461. Nonconformance to 3.31 shall constitute failure of the test.

4.4.15 Tiedown provisions test. The tiedown provisions shall be tested in accordance with MIL-STD-209 to prove conformance to 3.4.1. Inability to meet requirements of 3.4.1 shall constitute failure of this demonstration.

4.4.16 Slinging provisions test. The slinging provisions shall be tested in accordance with MIL-STD-209 to prove conformance to 3.4.2. Inability to meet the requirements of 3.4.2 shall constitute failure of this demonstration.

4.4.17 Air transport test. The trucks shall be tested in accordance with MIL-STD-1791 to prove conformance to 3.4.3. Inability to meet the requirements of 3.4.3 shall constitute failure of this demonstration.

4.4.18 Rail impact test. The first article trucks shall be tested in accordance with MIL-STD-810 to prove conformance to 3.4.4. Inability to meet the requirements of 3.4.4 shall constitute failure of this demonstration.

4.4.18.1 Failure criteria. Failure to meet the requirements of 3.4 and 4.4.18 rail impact test shall constitute failure of this test.

4.4.19 Highway transport test. Inability to meet the requirements of 3.4.5 shall constitute failure of this demonstration.

4.4.20 Drift. Test the forklift in accordance with test method No. 2, nonconformance to 3.24.13 shall constitute failure of this test.

4.5 Quality conformance. Each forklift shall be subjected to column 2, table 1, column 2, table II, and carrying a capacity load shall be subjected to a production run-in and servicing as specified herein. Inability to complete the run-in, overheating, failure of any components, malfunction of any control, or evidence of leakage of fluids shall be cause for rejection of the forklift by the Government until such deficiencies are corrected.

4.5.1 Production run-in. Demonstrate starting, ascending, and descending a 15 percent ramp, steering, and high speed travel. Operate all lights, horn, and brake system not less than two times during this demonstration.

4.5.2 Servicing and adjustment. Prior to delivery, the manufacturer shall service and adjust each forklift for customer operational use, including at least the following:

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a. Adjustment of mast assembly.
b. Adjustment of solid state control modules.
c. Adjustment of brake system.
d. Adjustment of steering.
e. Continuity check of the electrical system, including filling and charging the battery.

4.6 Inspection of packing. The preservation, packing, and marking of each forklift truck shall be inspected as specified in appendix B of MIL-STD-162 and as specified in 5.2.2.

4.7 Nuclear-biological-chemical (NBC) protection. The forklift shall be tested in accordance with 3.20.3. Nonconformance to 3.20.3 shall constitute failure of this test. The forklift shall complete two cycles in 7 minutes in accordance with test method no. 1 utilizing the NBC protective ensemble (see 3.20.3).

5. PACKAGING

5.1 Preservation. Each complete truck and all items issued with the truck shall be preserved in accordance with the level A, B, or C requirements for type I or II of MIL-STD-162, as specified (see 6.2).

5.2 Packing. Packing shall be level A, B, or C as specified (see. 6.2).

5.2.1 Level A or B. Each complete truck and all items issued with the truck, preserved as specified in 5.1, shall be packed in accordance with the level A or B requirements, type I or II, of MIL-STD-162.

5.2.2 Level C. Each complete truck and all items issued with the truck, preserved as specified in 5.1, shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification Rules or National Motor Freight Classification Rules.

5.3. Marking. Marking shall be in accordance with MIL-STD-162.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Trucks described herein are intended for stacking, unstacking, and moving cargo in and around warehouses, ammunition magazines loading platforms, and docks within the military supply system; also for moving cargo in and out of highway trailers and railroad cars. Trucks are intended for operation over paved, and other hard surfaces for short distances.

6.2 Acquisition requirements. Acquisition documents must specify the following:

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a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).
c. When first article test is not required (see 3.1.1).
d. When rail impact test is required (see 3.4).
e. When other than type E truck is required (see 3.2) .
f. When solid fill puncture proof tires are required (see 3.14).
g. When slinging and tiedown eyes are not required (see 3.4).
h. When batteries shall be shipped other than wet and charged (see 3.19).
i. When sideshift is not required (see 3.22).
j. When final color is other than manufacture's color (see 3.25).
k. When final color shall be 13538 of FED-STD- 595, (see 3.25).
l. When CARC is required (see 3.25).
m. When NBC protection is required (see 3.28).
n. When EMP testing is required (see 3.28).
o. When information and drawings are required for EMP testing (see 3.27).
p. When truck is required for EMP evaluation and testing (see 3.27).
q. When other than SB type battery connector is required (see 3.19.2).
r. Number of forklifts required for first article test and evaluation (see 4.3).
s. Level of preservation and packing required (see 5.1 and 5.2).
t. When electromagnetic interference is required (see 3.32).

6.3 First article. When a first article inspection is required, the item shall be a first produced forklift. The first article should consist of 1 or more units (see 6.2). The contracting officer should include specified instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the first article report.

6.4 Government-loaned equipment. The contracting officer should arrange to loan the equipment specified in 3.28.

6.5 Transportability report. When specified (see 6.2), the contracting officer should arrange for the contractor to furnish a transportability report in accordance with AR 70-47 and S.O.P 70-38 for the trucks, 95 days prior to delivery of the first article truck.

6.6 Subject terms (key word) listing.

Battery powered
Depot type forklift truck
Fork lift truck
Industrial forklift
Material handling equipment

6.7 Change from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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6.8 AQL certification. This specification is certified to be in compliance with current Army Materiel Command (AMC) policy for the elimination of AQL's/LTPD's (Acceptable Quality Levels/ Lot Tolerances Percent Defectives) from military specifications.

(Certified by/date: _____).

Custodian:

Army - ME
Navy - SA
Air Force - 99

Preparing activity:

Army - ME

Project 3930-0636

Review activities:

Army - AT, SM, MT
Navy - MC
DLA - CS
Air Force - 84

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FOR

TRUCKS, LIFT, FORK, ELECTRIC MOTOR DRIVEN,

2000-3000-4000-6000 POUND CAPACITY

10. SCOPE

10.1 Scope. The test methods contained in this appendix are to determine whether forklifts procured under this specification conform to the requirements set forth.

20. APPLICABLE DOCUMENTS

ASSOCATION OF AMERICAN RAILROADS (AAR)

Section 6 Rules Governing the Loading of Department of Defense Material on Open Top Cars

(Application for copies should be addressed to the Association of American Railroads, 50 F Street NW, Suite 7702, Washington, DC 20001.)

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TEST METHOD NO. 1

RELIABILITY TEST

1. Test course:

   a. Layout of course. The test course shall be set up equivalent to the provisions of figure 1. The course perimeter shall be demarcated with suitable barriers or indicators. Rubber pylons or other suitable marking shall be placed on all corners and other critical points to assist in keeping the truck being tested within the course limits.
   b. Obstacle inclusion. The obstacle test shall be set up in accordance with the provisions of figure 2 or figure 3 as applicable, as a part of the test course. Iron, wood, cement, or steel blocks may be used. A guideline 15 feet long shall be provided to the left of the centerline of the course and parallel to it. The inner edge of this guideline shall be at a distance from the course centerline equal to one-half of the truck width or one-half of the load width (whichever is greater) plus 6 inches. This spacing and arrangement will allow the left wheels of the truck under test to pass over the first block and than the right wheels to pass over the second block while the truck is moving forward in a straight line parallel to the guideline.
   c. Ramp inclusion. The course shall include a ramp constructed in accordance with the provisions of figure 4.
   d. Course surface. The course shall be paved with concrete having a fine float finish, or equivalent, having a road resistance of approximately 30-50 pounds per ton. It shall be dry, clean, and free of any nonplanned obstacles or foreign material while conducting the reliability test.
   e. Pallet areas. Pallet loads in the first pallet area shall be stacked in accordance with figure 5 or figure 6 as applicable. They shall be placed 2 inches apart in a direction parallel to, and touching in a direction perpendicular to the axis of the test course at that point. Side pallet stacks shall be high enough that the bottom of the top face of the top pallet in each stack is 6 inches below the maximum lift height of the vehicle under test. The stack marked "X" on figure 1 shall be high enough that the bottom of the top face of the pallet, when placed upon it, shall be 6 inches below the maximum lift height of the vehicle under test. Pallet loads in the second pallet area shall be stacked in accordance with figure 7 or figure 8 as applicable. They shall be placed 2 inches apart in a direction parallel to, and touching in a direction perpendicular to the axis of the test course at that point. Side pallet stacks shall be high enough so that the bottom of the top pallet will be approximately as high as the maximum lift height of the truck under test. The stack marked "Y" on figure 1 shall be high enough that the bottom of the top face of the pallet, when placed upon it, shall be at the midpoint between ground level and the maximum lift height of the vehicle under test. Pallet loads in the third pallet area shall be stacked in accordance with figure 9. They shall be placed 2 inches apart, positioned as shown on figure 1, in a direction parallel to, and touching in a direction perpendicular to the axis of the test course

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at that point. Side pallet stacks shall be one pallet load (48 inches plus the height of the pallet) high.

2. Test procedure:

Each truck shall complete 4,000 circuits of the test course as detailed below.

a. The light switch shall be turned "off" and "on" at the beginning of each lap and upon entering each pallet area.
b. The truck under test shall begin each lap at the point labeled "START" on figure 1. As the operator starts the truck, he shall operate the horn for approximately 1 second. On lap number 1, the truck shall proceed in a forward direction, following the centerline along the portion of the course marked "a" until it reaches the first pallet area. The rated load will have been placed in this location on the stack marked "X".
c. The truck shall make a 90-degree right-hand turn in one motion without backing, proceed to the face of the pallet stack marked "X", raise forks to maximum fork height so hydraulic relief valve is activated, then adjust fork height and remove the loaded pallet. The truck shall back out of the stack aisle into the main aisle, making a 90-degree left-hand turn in one motion, without going forward, and proceed rearward along the main aisle in the portion of the course marked "B" until it reaches the second pallet area.
d. The truck shall proceed rearward to a point beyond the pallet area opening, reverse its direction, and move forward to the face of the pallet stack marked "Y", making the 90-degres right-hand turn in one motion without backing. The rated load shall be placed on top of the pallet stack marked "Y".
e. The truck shall back out of the stack aisle into the rain aisle, making the 90-degree left-hand turn in one motion without going forward, and proceed forward along the main aisle in the portion of the course marked "C" until it reaches the third pallet area.
f. The truck shall rake a 90-degree right-hand turn in one motion without backing and pick up rated load which has been placed in the location marked "Z". The truck shall back out into the main aisle, making the 90-degree right-hand turn in one motion, and proceed further along the main aisle in the portion of the course marked "D" until it reaches the portion of the course marked "E", containing the obstacle blocks.
g. The obstacle blocks shall be traversed by the truck being driven through this area in a straight line so that its left wheels pass over the first block and the right wheels pass over the second block. This part of the test may be conducted at low speed with the transmission in low range. However, the truck shall not be stopped prior to or while going over the blocks.
h. The truck shall proceed further along the rain aisle in the portions of the course marked "F" and "G" until it reaches the position on the ramp labeled "Stopping Point". The driver shall stop the truck and then resume ascending the ramp. After the top of the ramp is reached, proceed along the main aisle in the portion of the course marked "J", "K", and "L".

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i. When the truck reaches the portion of the course marked "M", the operator shall stop. On every other lap, the operator shall turn off the master switch, then turn "ON". After completion of this portion of the test, proceed along the rain aisle and start the next cycle. All stops shall be at the maximum safe deceleration rate. The operator shall actuate and release the parking brake at least once during each circuit.

j. The truck shall then reenter the portion of the course marked "A". In portions "A", "B","C", and "D", the procedure shall be the same as described in preceding paragraphs, except that the truck shall place a test load at each point where it was previously indicated that the truck picked one up, and pick up a load whom it was previously indicated one was placed. Maneuvers in the balance of the course beyond the portion marked "D" shall be the same as described in preceding paragraphs.

k. One-half of the total number of laps shall be traversed in the opposite direction with all operation reversed accordingly. This may be accomplished after completion of each 8-hour operating day or multiple thereof. That is, 50 percent of the laps shall be accomplished in the clockwise direction and 50 percent in the counterclockwise direction or vice versa. The test course need not be rearranged for the reverse travel.

1. "The Hourly Time Record Sheet" shall be filled out for each operating hour of the performance test. Average lap speed per hour shall be recorded.

m. Maintenance and inspections shall be performed in accordance with the maintenance service and inspection sheet.

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TEST METHOD NO. 2

DRIFT

1. TEST COURSE:

   a. Level, flat surface.

2. TEST APPARATUS:

   a. Temperature thermocouple and potentiometer or equivalent temperature measuring device.

   b. Clinometer.

3. TEST PROCEDURE:

   a. Install the thermocouple in the volumetric center of the hydraulic-system oil reservoir. Heat the hydraulic oil by exercising hydraulic functions until the oil temperature exceeds 120 degrees Fahrenheit. Maintain oil temperature above 120 degrees Fahrenheit for the duration of the test.
   b. Raise rated load to maximum lift height. Tilt mast to true vertical position measured with the clinometer. Mark reference point.
   c. Hold the load for 2 minutes and record the downward drift and rotational drift in that period.

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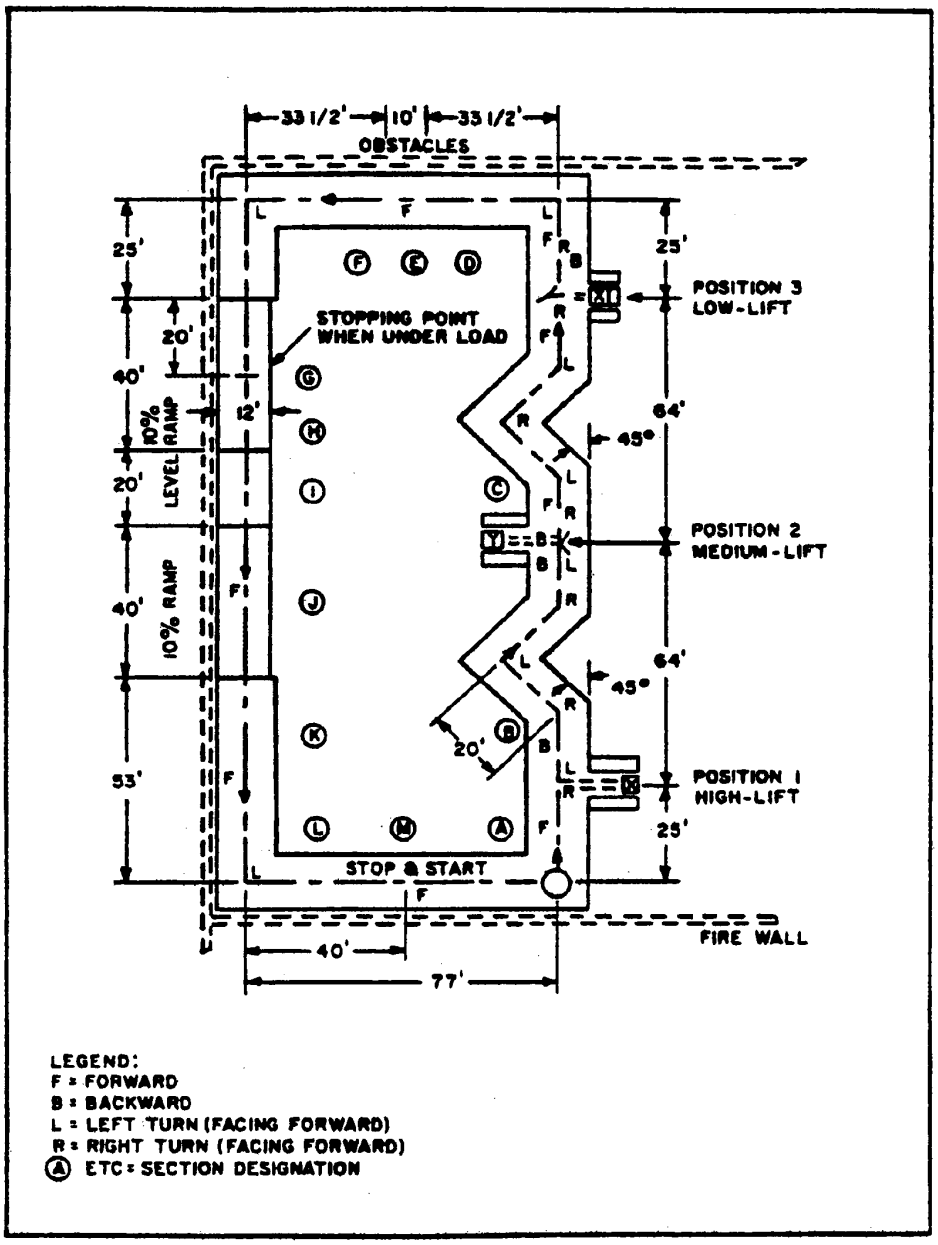


FIGURE 1. Typical indoor test course.

X-4494

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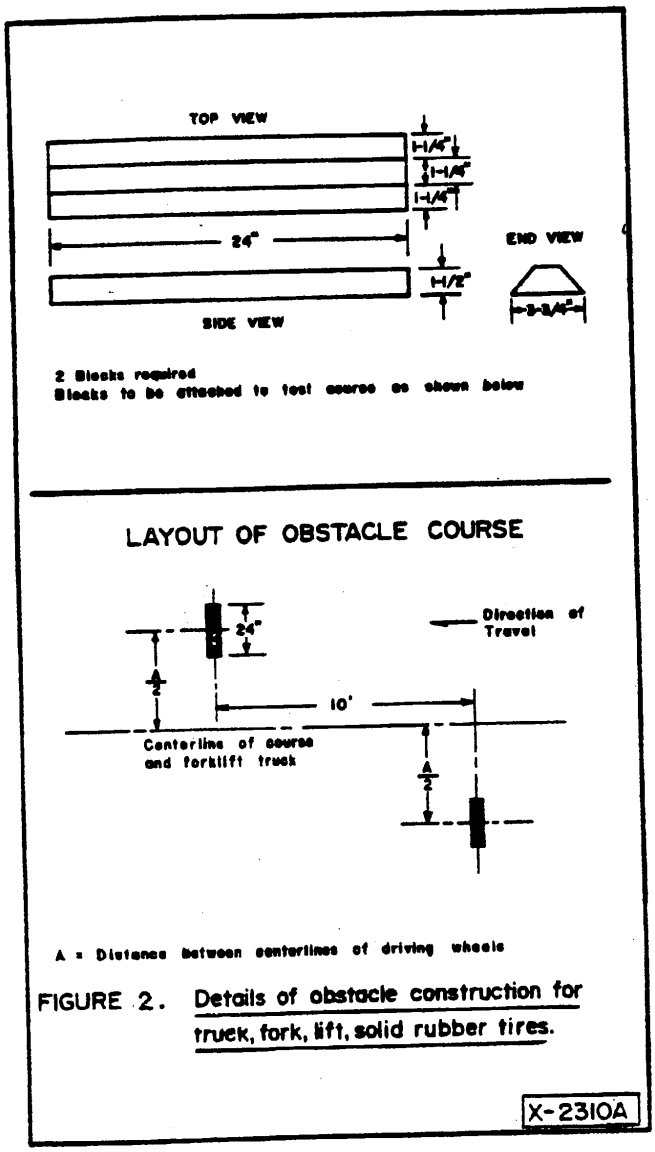
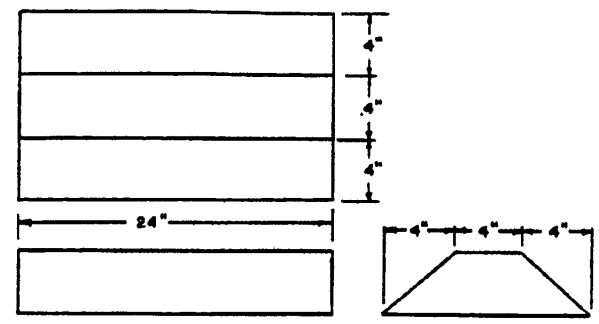


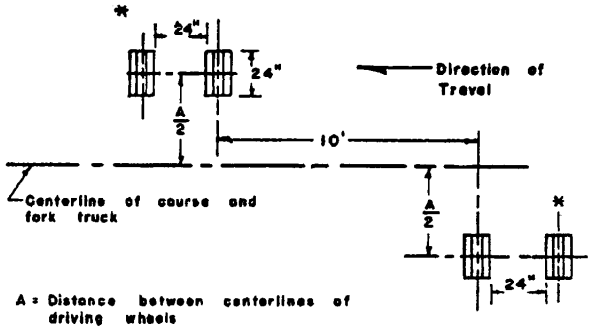
FIGURE 2. Details of obstacle construction for truck, fork, lift, solid rubber tires.

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HEIGHT OF OBSTACLE BLOCK
2" for 2000 and 4000 pound trucks
3" for 6000 pound trucks

LAYOUT OF OBSTACLE COURSE



A = Distance between centerlines of driving wheels

* Size 2000 pounds, pneumatic rubber tired trucks shall require only one test block on each side of the centerline.

Figure 3. DETAILS OF OBSTACLE CONSTRUCTION FOR TRUCK, FORK, LIFT, PNEUMATIC RUBBER TIRES.

X-2311A

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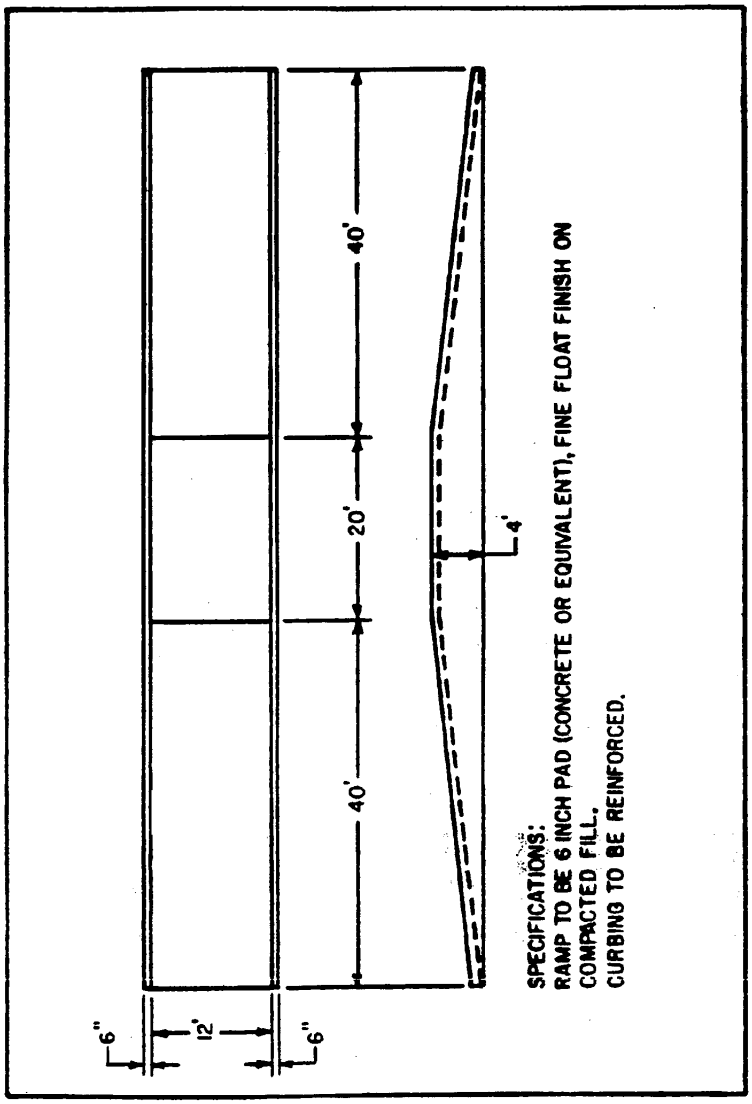
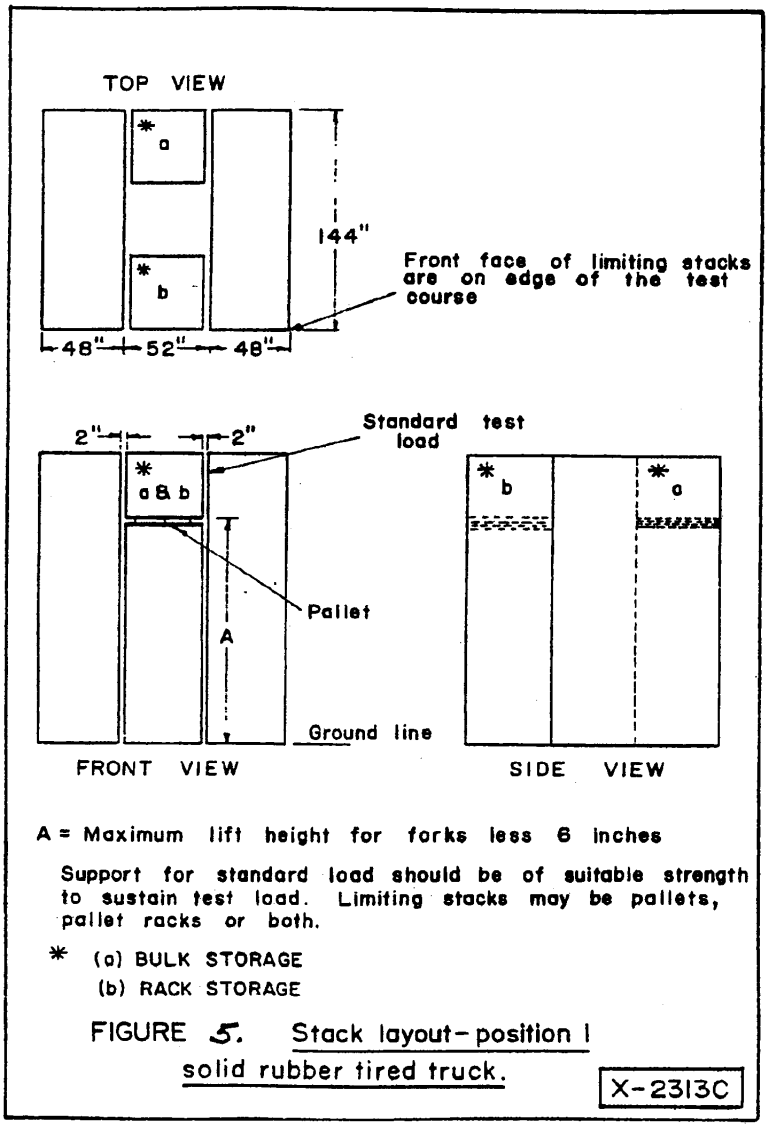


FIGURE 4. Ramp design.

X-4495

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A = Maximum lift height for forks less 6 inches

Support for standard load should be of suitable strength to sustain test load. Limiting stacks may be pallets, pallet racks or both.

* (a) BULK STORAGE
(b) RACK STORAGE

FIGURE 5. Stack layout - position 1 solid rubber tired truck.

X-2313C

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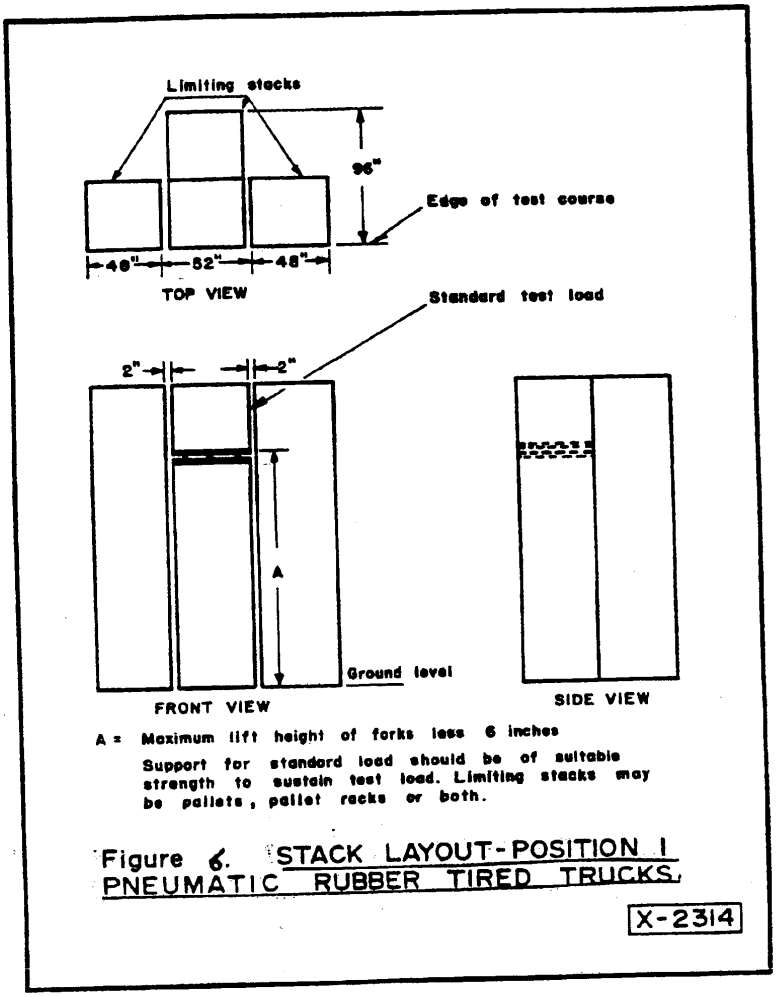
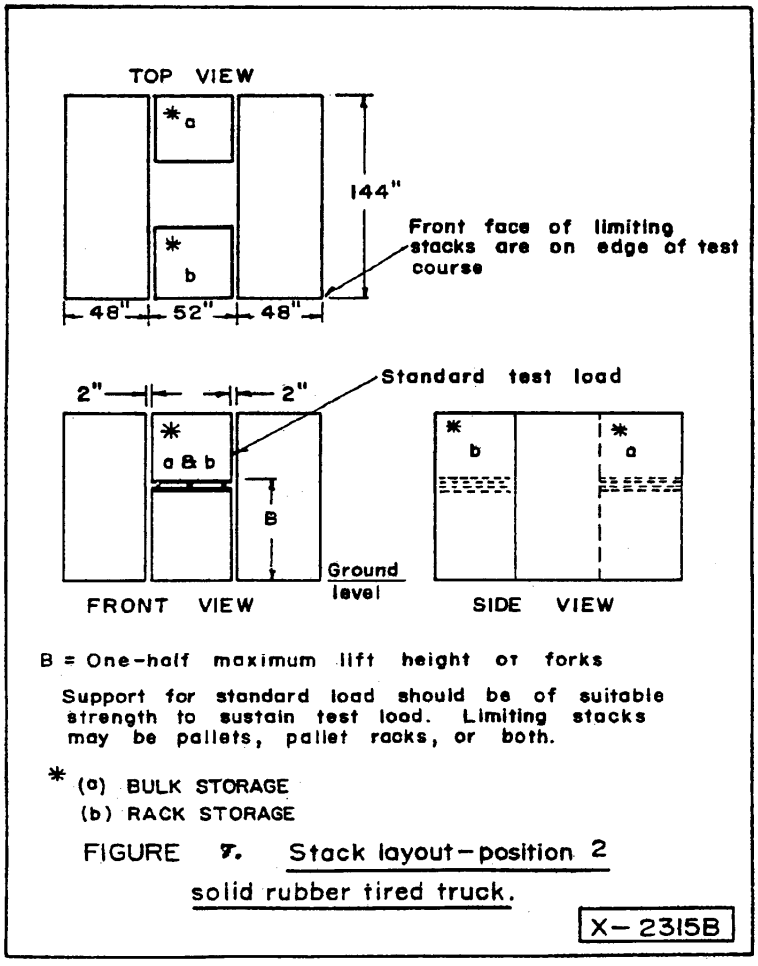


Figure 6. STACK LAYOUT-POSITION 1 PNEUMATIC RUBBER TIRED TRUCKS.

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B = One-half maximum lift height of forks

Support for standard load should be of suitable strength to sustain test load. Limiting stacks may be pallets, pallet racks, or both.

* (a) BULK STORAGE
(b) RACK STORAGE

FIGURE 7. Stack layout—position 2 solid rubber tired truck.

X-2315B

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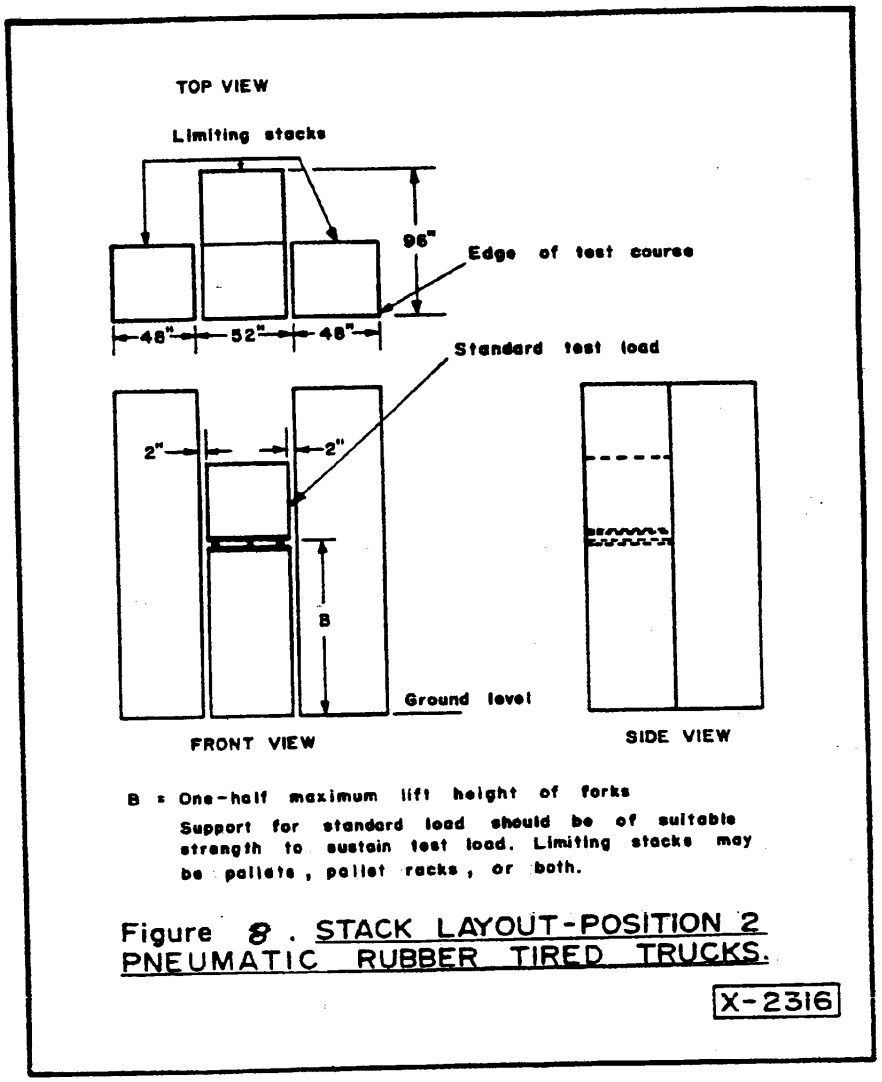


Figure 8. STACK LAYOUT-POSITION 2 PNEUMATIC RUBBER TIRED TRUCKS.

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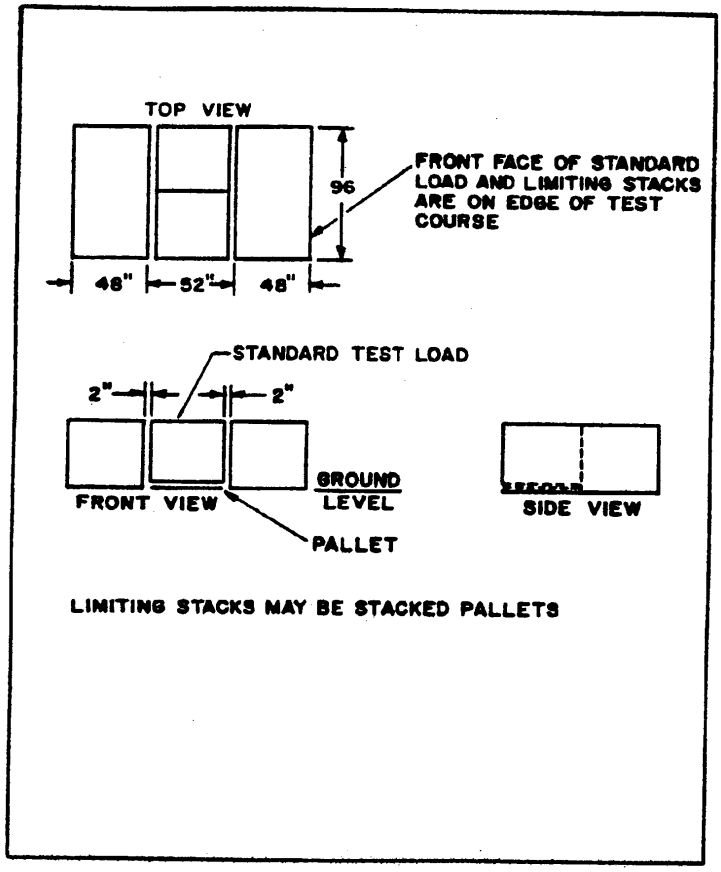


FIGURE 9. Stack layout - position 3.

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TEST PROCEDURE FOR RAIL SHIPMENT

Step 1. From one to five rail cars shall be positioned on a level section of track. The following conditions shall be satisfied:

a. The total weight of the car(s) shall be at least 114,000 kilograms (250,000 pounds).
b. The couplers between cars shall be compressed to take up any slack.
c. All of the air and hand brakes on the car(s) used shall be set.
d. Any load in or on the car(s) shall be secured to prevent sliding or shifting; any movement greater than 5 cm shall be justification for retest.
e. The end of the buffer car to be struck must have a standard draft gear.

Step 2. Mount the test item on the test car. The test car shall be equipped with standard draft gears and conventional underframes. Mounting of the test item shall incorporate the standard loading and bracing method as shown in section 6 of the Association of American Railroads (AAR) "Rules Governing the Loading of Department of Defense Materiel on Open Top Cars." No exotic or unusual tiedown methods shall be used; any non-standard loading and bracing must be approved by the Military Traffic Management Command Transportation Engineering Agency (MTMCTEA), ATTN: MTT-TR, P.0. Box 6276, Newport News, VA 23606-0276 prior to testing. Substitute (dummy) test items must be of equal weight and weight distribution as the actual test item and similar in shape. The arrangement of the test item and its blocking and tiedown to be tested shall be identical to that proposed and approved by MTMCTEA (if non-standard).

This requirement is mandatory for all equipment developed for use in the U.S. Equivalent European standards may apply for non-U.S. commodities.

Step 3. Situate the test car between the buffer cars and the locomotive, and pull the test car at least 65 meters (200 feet) from the buffer car(s) along a level section of track (a minimum distance to achieve the required locomotive speeds).

Step 4. Position the knuckles of the buffer and test cars for coupling.

Step 5. Install a timing device to measure the test car speed (± .1 km/hr) just prior to impact with the buffer car(s). Suggested methods include electronic timing (microswitches) and radar. The use of torpedoes and a stopwatch is permissible but not recommended because of the inaccuracies involved.

Step 6. Push the test car toward the buffer car(s) and, by using the locomotive's speedometer or other means, release the test car when the desired test speed is reached, thus allowing the test car to freely impact the buffer car (s).

Step 7. Repeat step 6 until the test car(s) has impacted the buffer car three times at the same end, once each at speeds of 6.4, 9.7 and 13.0 km/hr (4, 6,

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and 8 mph) +0.8, -0.0 km/hr. Reverse the test car and repeat the 13 .0 km/hr impact, for a total of 4 impacts.

NOTE

Adjustment of the lading or securing mechanisms, or reconditioning of the bracing or items of securement is not allowed without a complete retest.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter must be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-T-52953C	2. DOCUMENT DATE (YYMMDD) 891027

3. DOCUMENT TITLE
TRUCKS, LIFT, FORK, ELECTRIC MOTOR DRIVEN, 2000-3000-4000-6000 POUND CAPACITY GENERAL SPECIFICATION FOR

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code)) Commercial (2) DSN (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME USA TACOM, AMSTA-TR-E-ELE/21 (4EELD)	b. TELEPHONE (Including Area Code) (1) Commercial (810)574-6516	(2) DSN 786-6516
c. ADDRESS (Include Zip Code) US Army Tank-automotive and Armaments Command Attn: AMSTA-TR-E-ELE/21 Warren, MI 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 681-9340	DSN 761-9340

DD Form 1426, Oct 89 *Previous editions obsolete*

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